

Software Requirements Specification

Automatic Train Interlocking System

Table 1: Document information

Item	Value
System name	Automatic Train Interlocking System
Document type	SRS
Main scope	Safety logic, track occupancy, route reservation, and train movement simulation
Main input	Legal XML railway layout file
Excluded scope	Timetable planning, train speed modelling, and high-level service planning

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1. Project Overview

The Automatic Train Interlocking System is an academic software prototype for loading a railway layout from XML, configuring track occupancy, checking whether a safe route exists between a source and destination, reserving the route, and simulating train movement along it.

The system is strictly focused on interlocking-style safety logic. It does not model timetable planning, train speed, passenger service planning, or traffic optimization.

2. Scope

Table 2: Scope of the system

In scope	Out of scope
Load legal XML railway layout files.	Timetable planning or schedule optimization.
Parse track sections, points/switches, markerboards/signals, and neighbor links.	Train speed, acceleration, braking curves, or detailed physics.
Allow user to configure initial track occupancy.	Passenger service planning or commercial train service data.
Accept source track and destination track service requests.	Real certified safety-critical deployment.
Find a valid free and unreserved route.	Authentication, authorization, or external network deployment.
Reserve all track sections and points in the route.	Full industrial signalling rulebook implementation.
Simulate train movement and update occupancy state.	Automatic timetable recovery after disruptions.
Persist current occupancy state across iterative requests.	Permanent long-term storage of every live state unless later added.
Allow clear state or load new layout.	Multi-user traffic control center operation.

3. Users and System Context

Table 3: Users and responsibilities

User	Responsibility
Scenario user	Loads layout, defines initial occupancy, enters source/destination, and runs simulation.
Developer/student	Implements, tests, explains, and maintains the prototype.
Reviewer	Checks requirements, workflow, diagrams, and whether the implementation matches the SRS.

4. Functional Requirements

Table 4: Functional requirements

ID	Requirement	Priority
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FR-01	The system shall read and load any legal XML layout file following the expected layout structure.	High
FR-02	The system shall parse track sections, points/switches, markerboards/signals, and neighbour connectivity from the loaded XML layout.	High
FR-03	The system shall build an internal topology graph from the loaded layout.	High
FR-04	The user shall be able to configure initial track occupancy on the loaded layout.	High
FR-05	The user shall be able to define a service request with a source track and destination track.	High
FR-06	The system shall determine whether a valid connected route exists between source and destination.	High
FR-07	The route shall use only unreserved and free track sections, except that the source track already occupied by the requesting train is not blocked by its own occupancy.	High
FR-08	The route shall include required points/switches as reserved resources.	High
FR-09	If a valid route is found, the system shall reserve all track sections and points required by the route.	High
FR-10	If no valid route exists, the system shall reject the request and show a reason.	High
FR-11	If a route is reserved, the system shall simulate train movement along the route.	High
FR-12	During simulation, the system shall dynamically update the occupancy of track sections.	High
FR-13	When simulation finishes, the updated occupancy state shall persist for further requests.	High
FR-14	If route finding fails, the current occupancy state shall persist unchanged.	High
FR-15	The user shall be able to issue additional service requests iteratively on the same persisted state.	High
FR-16	The user shall be able to clear the state and define a new scenario on the same layout.	High
FR-17	The user shall be able to load a new XML layout file, replacing the previous layout and state.	High

FR-18	If the supplied XML layout file is not legal or well-formed, the system shall reject it and report a clear error instead of loading a partial or corrupt layout.	High
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5. Non-Functional Requirements

Table 5: Non-functional requirements

ID	Requirement	Explanation
NFR-01	Correctness	A route must not be accepted if it contains occupied or reserved resources.
NFR-02	Traceability	The system should show why a route was accepted or rejected.
NFR-03	Usability	The dashboard should clearly show layout loading, occupancy, route request, route result, and current state.
NFR-04	Testability	Parser, route finding, reservation, simulation, state persistence, clear, and reload behavior should be testable.
NFR-05	Maintainability	Layout parsing, route finding, state management, and simulation should be separated into clear components.
NFR-06	Safety-first behavior	Safety checks must happen before reservation or movement simulation.

6. Topology Definition

Topology is the railway layout loaded from XML. It describes the resources and connections used by route finding.

Table 6: Topology terminology

Term	Meaning in this system
Track section	A physical track resource that can be free, occupied, or reserved.
Point / switch	A branching track resource that must be reserved together with the route.
Markerboard / signal	A layout element that can identify route boundaries or be mapped to track sections.
Neighbor link	A connection between two track sections.
Route	A connected chain of track sections and points from source to destination.
Occupancy state	The current position of trains on the layout.
Reservation state	The resources temporarily locked for a route.

In the user interface, the source and destination may be selected using track section IDs or markerboards mapped to track sections. Internally, route finding works on the topology graph.

7. State Persistence Rules

Table 7: State persistence rules

Rule	Required behavior
SPR-01	The current occupancy state remains available for further service requests until the user explicitly clears it or loads a new XML layout.
SPR-02	After a successful simulation, the train's movement updates the occupied/free state of the affected track sections.
SPR-03	After route finding failure, no train movement occurs and the current occupancy state remains unchanged.
SPR-04	A new service request uses the current persisted state, not the original empty layout.
SPR-05	Clear state removes scenario occupancy/reservations but keeps the loaded layout.
SPR-06	Loading a new layout replaces both the previous layout and the previous state.

8. Constraints and Assumptions

Table 8: Constraints and assumptions

Area	Statement
Prototype status	The system is an academic prototype and is not certified railway safety software.
XML input	The system expects legal XML layout files according to the supported structure.
Safety focus	The system focuses on occupancy, reservation, route availability, and movement state.
No speed model	The system does not calculate train speed, acceleration, braking distance, or running time.
No timetable model	The system does not manage timetable planning or service-level scheduling.
Single-user prototype	The prototype assumes one scenario user operating the dashboard at a time.
State lifetime	State persistence is required during the active loaded layout/session until clear or reload.

9. Main Workflow

1. The user loads a legal XML railway layout.
2. The system parses track sections, points/switches, markerboards/signals, and neighbor links.
3. The user configures initial occupancy by placing trains on selected track sections.

4. The user submits a service request using a source track and destination track.
5. The system searches the current topology graph for a connected route.
6. The route search rejects occupied or already reserved sections.
7. If no valid route exists, the request is rejected and the current state remains unchanged.
8. If a valid route exists, the system reserves all sections and points in the route.
9. The system simulates train movement along the route.
10. The system updates occupancy dynamically during movement.
11. The final occupancy state persists for additional service requests.
12. The user may continue, clear the state, or load a new XML layout.

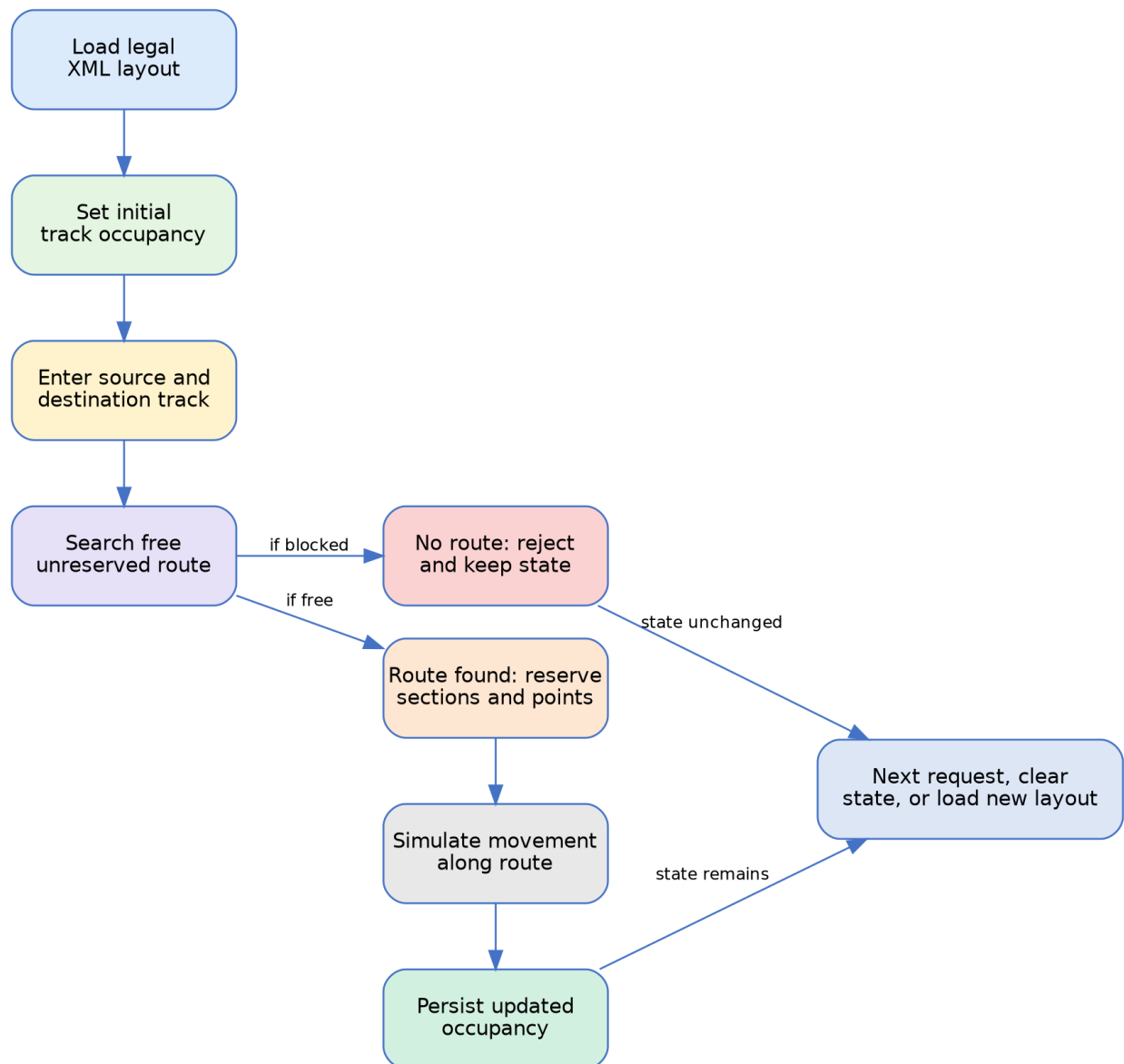


Figure 1: Activity diagram - route reservation and simulation

10. Main Components

Table 9: Main components and responsibilities

Component	Responsibility
XML Layout Parser	Reads the legal XML layout and extracts topology resources.
Layout Graph	Stores track sections and neighbor connectivity.
Scenario State Manager	Stores current occupancy and reservation state.
Route Finder	Finds a connected route between source and destination using only free/unreserved resources.
Route Reservation Logic	Locks all track sections and points required by the selected route.
Movement Simulator	Moves the train along the route and updates occupancy dynamically.
Dashboard/API	Supports load layout, configure scenario, request route, run simulation, clear state, and reload layout.

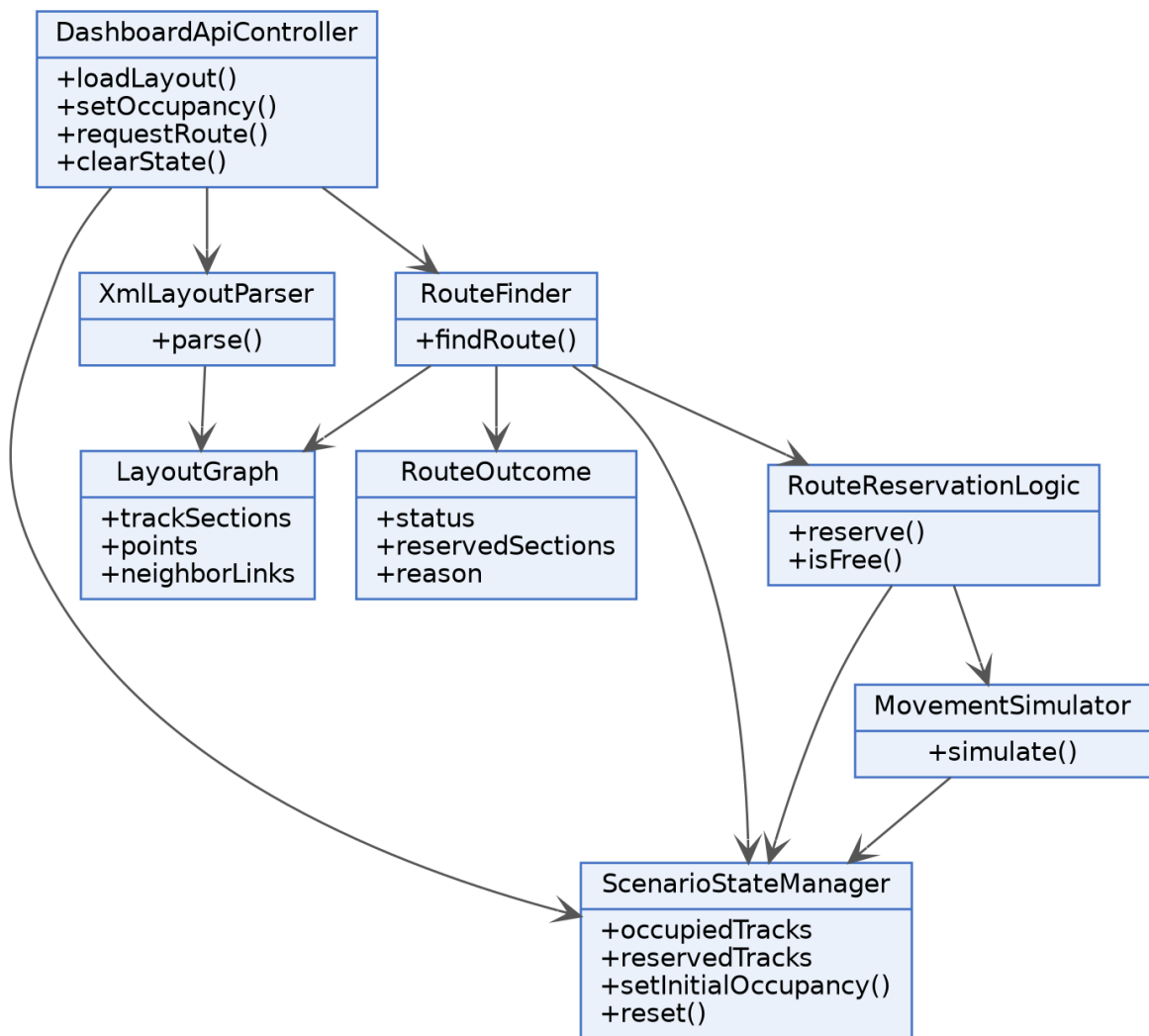


Figure 2: Class diagram - core domain classes

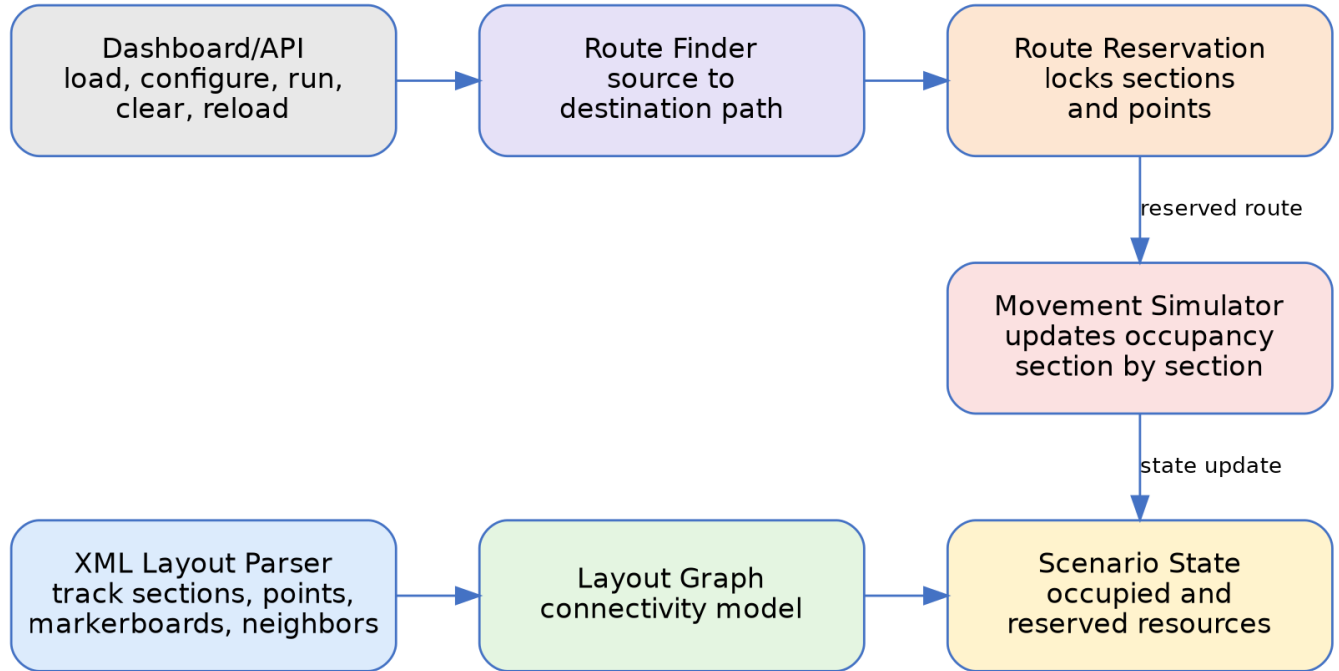


Figure 3: Component diagram - focused interlocking scope

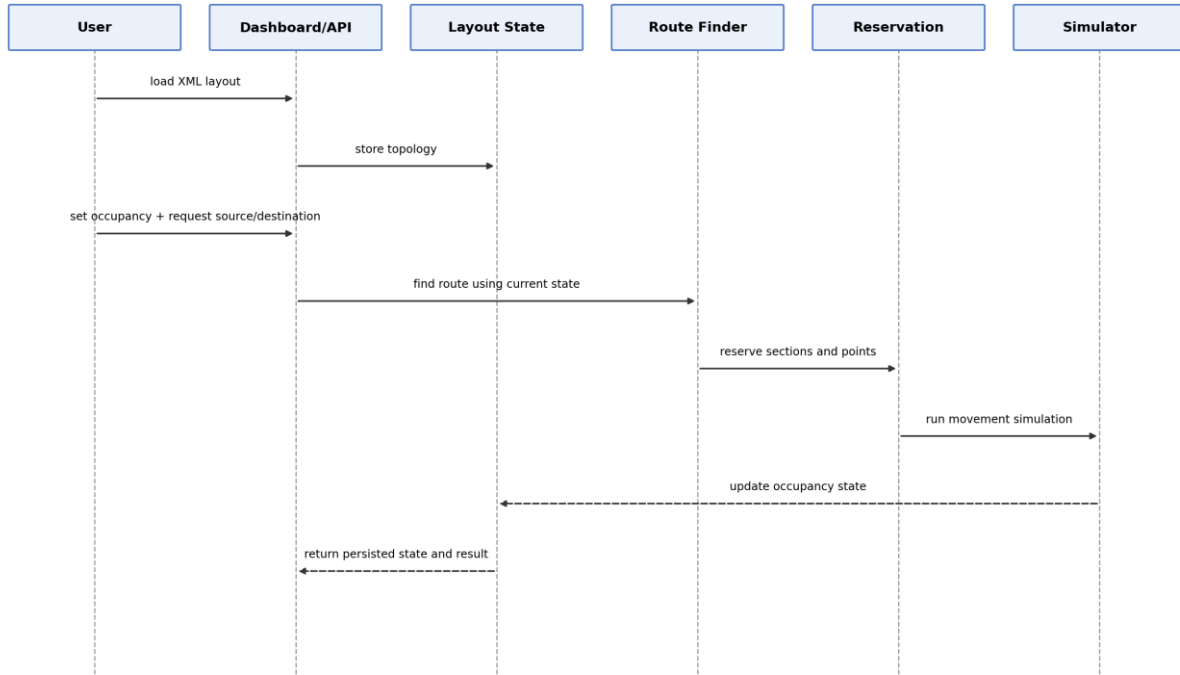


Figure 4: Sequence diagram - one service request

11. Acceptance Criteria

Table 10: Acceptance criteria

ID	Acceptance check
AC-01	A legal XML layout can be loaded successfully.
AC-02	The loaded layout produces an internal topology graph.
AC-03	The user can set initial occupancy on selected track sections.
AC-04	The user can submit source and destination track values.
AC-05	The system accepts a request only when a free unreserved route exists.
AC-06	The system rejects a request when every possible route is blocked.
AC-07	A successful route reserves all route sections and points.
AC-08	The movement simulation updates occupancy section by section.
AC-09	The updated occupancy state remains for the next request.
AC-10	Failed route finding does not reset or corrupt the existing state.
AC-11	Clear state starts a new scenario on the same layout.
AC-12	Loading a new XML layout replaces the previous layout and state.
AC-13	Loading a malformed or illegal XML layout file is rejected

	with a clear error message, and no layout/state change occurs.
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12. Implementation Notes

The implementation should keep layout parsing, route finding, reservation, state management, and movement simulation as separate units. This makes the safety logic easier to test and explain.

Audit/history may be kept as supporting information, but the main thesis scope is the current interlocking state: which track sections are free, occupied, or reserved.

13. AI Tools Declaration

AI tools were used as support for drafting, reorganizing, wording, and preparing intermediate documentation ideas. The final content was reviewed and adjusted by the student. AI assistance does not replace implementation, testing, or responsibility for the submitted work.